

COMPLICATIONS IN PEOPLE LIVING WITH TYPE 1 DIABETES 15 YEARS POST DIAGNOSIS BY DECADE OF DIAGNOSIS : A CROSS-SECTIONAL, REAL-WORLD OBSERVATIONAL STUDY

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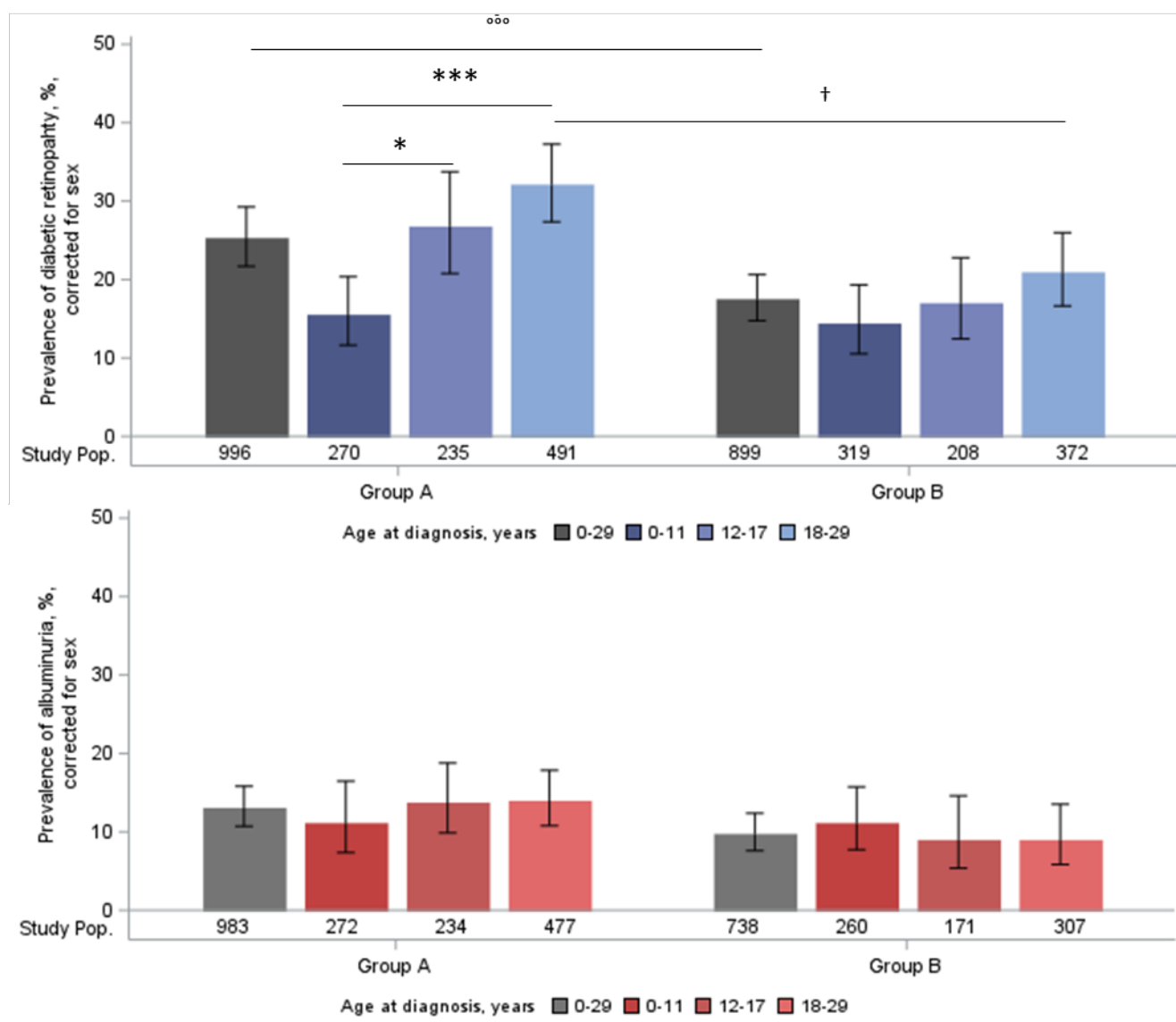
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Background: Advances in blood lipid control and glucose management have reduced complications and mortality in type 1 diabetes. This study examined the prevalence of diabetic retinopathy and albuminuria 15 years post-diagnosis in individuals diagnosed before and after 1998, alongside mortality rates.

Methods: Data were sourced from the Initiative for Quality Improvement and Epidemiology in Diabetes. Group A included individuals diagnosed between 1985-1998, with their 15-year post-diagnosis period falling between 2001 and 2011. Group B included individuals diagnosed between 1998-2009, with their 15-year post-diagnosis period occurring between 2012 and 2022. All were <30 years at diagnosis, categorized into three age groups: 0–11, 12–17, and 18–29 years. Generalized Estimating Equations analyzed complication prevalence, and Kaplan-Meier methods assessed cumulative mortality. Cox regression determined associations between patient characteristics and mortality risk.

Results: At 15 years post-diagnosis, Group A was older (32. vs. 31.6 years, $p<0.001$), had a shorter diabetes duration (14.9 vs. 15.2 years, $p<0.0001$), and a higher age at diabetes onset (17.8 vs. 16.4 years, $p<0.0001$). Mean HbA1c (67.4 vs. 64.6 mmol/mol, $p<0.0001$) and LDL cholesterol (2.6 vs. 2.5 mmol/l, $p<0.001$) were higher in Group A, while obesity was more common in Group B (17.4% vs. 12.8%, $p<0.01$). Antihypertensive therapy use was greater in Group A (14.9% vs 11.5%, $p<0.05$), while lipid-lowering therapy (15.9% vs 9.2%, $p<0.0001$) and insulin pump use (13.8% vs. 1.0%, $p<0.0001$) were more common in Group B. Prevalence of retinopathy and albuminuria are shown in figure. After 10 years, mortality incidence was lower in Group B (2.1% vs. 5.5%, $p<0.05$), with a hazard ratio of 0.423 (95% CI 0.213–0.84, $p=0.0141$).

Conclusions: Diabetic retinopathy and albuminuria prevalence declined in those diagnosed post-1998, and especially for retinopathy at older onset ages. Survival rates were better for those diagnosed post-1998, independent of age at onset and sex.



* $p < 0.05$; *** $p < 0.001$, between age at diagnosis categories, per study group
† $p < 0.05$, between study groups, per age at diagnosis category
††† $p < 0.001$, between study groups (corrected for age at diagnosis and sex)